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Class:- Sy-3 ; C3

Subject:- DSA

Exp no.:- 6

Title: Implement Stack using Linked List.

*Faqs:-

1. What is a Queue?

→ A queue is a linear data structure that follows the FIFO (First In, First Out) principle, where elements are inserted at the rear end and deleted from the front end.

2. How is a Queue represented?

→ A queue can be represented sequentially using an Array or dynamically using a Linked List.

3. Enlist types of Queues.

→Linear Queue (Simple Queue)

Circular Queue

Priority Queue

Double-Ended Queue (Deque)

4. What is the concept of a Circular Queue?

→A circular queue is a linear data structure where the last position is connected back to the first position to form a circle, overcoming memory wastage issues in simple linear queues.

5. What are the applications of a Queue?

→1.CPU & Disk Scheduling: Managing processes waiting for resource allocation.

2. Data Buffering: Handling asynchronous data transfer (e.g., IO buffers, printer queues).

3. Queue Simulation: Modeling real-world waiting lines (e.g., call centers, traffic control).

6. What is a Priority Queue?

→ A priority queue is a specialized queue where each element has an associated priority:

- Elements with higher priority are served before lower priority elements.
- Elements with equal priority are served on a FIFO basis.

7. What is a Deque?

→ Deque (Double-Ended Queue) is a linear data structure where elements can be inserted or deleted from **both** the front and rear ends, but not from the middle.

Exp no.:- 7

Title: Evaluation of postfix expression.

*Faqs:-

1. What are the different types of expressions?

   → Expressions are categorized based on where the operator is placed relative to its operands. In Infix, the operator sits between operands (a + b). In Prefix (Polish Notation), the operator comes before (+ a b). In Postfix (Reverse Polish Notation), the operator comes after (a b +).

2. Give examples of all types of expressions.

   →Infix: a + b (Standard arithmetic format used by humans).

   Prefix: + a b (Operator precedes operands).

   Postfix: a b + (Operator follows operands, making it easier for machines to evaluate without parentheses).

3. Which data structure is used for evaluation of expression?

   →A Stack operates on a Last-In, First-Out (LIFO) principle. This allows it to store operands and operators in order, popping them to evaluate sub-expressions according to operator precedence and parentheses rules.

4. Convert infix expression a+b+c+d to prefix expression.

   →Left-associative operators (+) of equal precedence are evaluated from left to right:

1. Convert (a + b) → +ab
2. Convert ((+ab) + c) → ++abc
3. Convert (((++abc)) + d) → +++abcd

5. Convert infix expression a+b+c+d to postfix expression.

→Evaluating left to right:

1. Convert (a + b) → ab+
2. Convert ((ab+) + c) → ab+c+
3. Convert (((ab+c+)) + d) → ab+c+d+

Exp no.:- 8

Title: Implement Binary search tree with operations Create, search, and recursive traversal.
*Faqs:- No questions available in the manual

Exp no.:- 9

Title: Represent graphs and generate minimum spanning trees using Prim's algorithm.
*Faqs:-

1. What is a tree?

   →A tree is a non-linear, hierarchical data structure consisting of nodes connected by edges, starting from a single root node.

2. What is a binary tree?

   →A binary tree is a tree data structure in which each node has at most two children, referred to as the left child and the right child.

3. What are the types of binary tree?

   →Full Binary Tree, Complete Binary Tree, Perfect Binary Tree, Balanced Binary Tree, Skewed Binary Tree, Binary Search Tree (BST), and Threaded Binary Tree.

4. What are tree traversal techniques?

   →Depth-First Traversal (Recursive / Non-recursive using Stack)

   Breadth-First Traversal (Level Order using Queue)

5. What are recursive tree traversal techniques?

   →Preorder: Root → Left → Right

   Inorder: Left → Root → Right
   Postorder: Left → Right → Root

6. What are AVL trees?

   →An AVL tree is a self-balancing binary search tree where the height difference (balance factor) between the left and right subtrees of any node is at most 1.

7. What are the basic operations that can be performed with a tree?

→ Insertion: Add a node.

Deletion: Remove a node.

Search: Find a specific node.

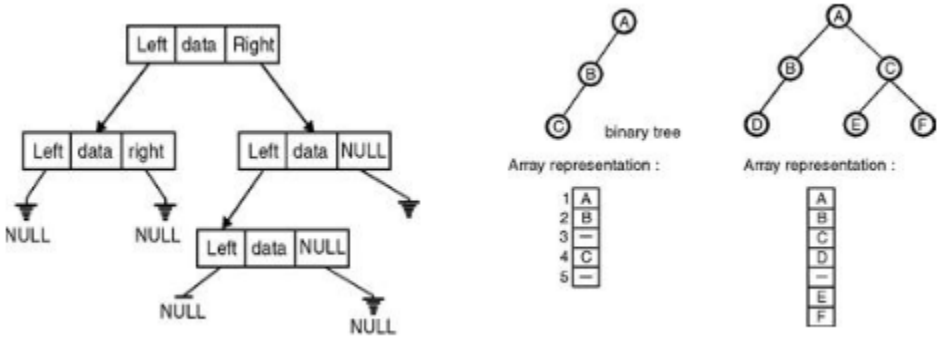
Traversal: Visit all nodes in a specific order.

8. Explain the terms degree and height of a tree.

   →Degree of a node/tree: The degree of a node is its number of children. The degree of the tree is the maximum degree among all its nodes.

   Height of a tree: The length of the longest path from the root node to a leaf node (counted in edges or levels).

9. Draw representation of tree using Linked List and Array.



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Exp no.:- 10

Title: Implement UART communication using Data Structures in C

*Faqs:-

1. What is UART?

   → UART (Universal Asynchronous Receiver-Transmitter) is a hardware communication protocol that transmits data serially (one bit at a time) between two devices using only two signal lines: transmit (TX) and receive (RX).

2. Why use circular buffer for UART?

   → A circular buffer acts as a ring-like memory queue for incoming or outgoing bytes. It allows data to be read and written continuously in fixed memory using head/tail pointers, preventing

memory shifting and preventing data loss when UART receives data faster than the main program processes it.

3. What happens when buffer is full?

   → When the head pointer catches up to the tail pointer, the buffer has no space left. New incoming bytes will either overwrite older unread data (data corruption/loss) or be discarded until the reader consumes existing data to free up space.

4. Is UART synchronous or asynchronous?

   → UART is **asynchronous**. It does not use a shared clock signal line. Instead, both the sender and receiver agree on a fixed timing speed (baud rate) beforehand and synchronize using start and stop bits added to each byte.

5. What is baud rate?

   → Baud rate is the rate at which data is transferred over a serial line, expressed in bits per second (bps). Common standard baud rates include 9600 bps and 115200 bps; both transmitting and receiving devices must set the exact same baud rate to decode bytes correctly.